

QUOC HUY LIEU

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SUMMARY

Hardware engineer with hands-on PCB design, embedded firmware, and hardware debug experience across automotive ECU validation and FPGA-based hardware security research. Designed and laid out a PCB in Altium, used JTAG, oscilloscope, and logic-analyzer methods to isolate hardware-level defects, and built bare-metal embedded firmware on ARM Cortex-M4. Comfortable working directly with schematics, firmware, and system-level test to own problems end-to-end.

TECHNICAL SKILLS

Debug & Bench Instruments: Oscilloscope, Logic Analyzer, Multimeter (DMM), JTAG Debugging, Serial Debugging (RealTerm)

Embedded Systems & Firmware: Bare-Metal C/C++ (ARM Cortex-M4), RTOS Design, SPI, I2C, UART, FPGA Synthesis & Implementation, RTL (Verilog)

Hardware Design: PCB Design & Layout (Altium Designer), Schematic Capture & Reading, Component Selection, Datasheet Analysis

Validation & Test: HIL Testing (Vector CANoe, VT System), CAN/CAN-FD, UDS (ISO 14229), Failure Reaction & DTC Analysis, Requirements Traceability

Programming & Automation: Python, C, C++, TCL, CAPL, VBA

Tools: Vivado, ChipWhisperer, VTestStudio, WinIDEA, Git, JIRA, IBM DOORS

PROFESSIONAL EXPERIENCE

Bosch Global Software Technologies Vietnam

Ho Chi Minh City, Vietnam

Embedded System Test Engineer

Dec 2021 – Aug 2023

- Identified 20+ signal-level defects before vehicle integration and prevented costly system-level rework by validating ADAS features (PAS, PSL, SDW) on SPC5 (PowerPC) MCUs in HIL environments using Vector CANoe and VT System.
- Used JTAG debugging alongside vehicle and sensor geometry analysis to investigate PAS HMI internal behavior, expanding coverage across positive, negative, and configuration-dependent cases and converting major scenarios from manual to automated testing.
- Exposed 5+ DTC defects across PAS, PSL, and SDW by designing fault injection sequences targeting fail-safe and heal-path transitions not covered by existing test plans.
- Cut per-session diagnostic setup time 50% and eliminated data-entry errors by building a CAPL-driven GUI diagnostic panel in CANoe that replaced manual UDS message entry; adopted as the team standard workflow.
- Reduced diagnostic defects 60% within 6 months by designing an ISO 14229-compliant UDS diagnostic template covering 50+ test cases, including previously absent negative and boundary conditions.
- Enabled 3 interns and 2 newcomers to independently execute assigned test features within 2 weeks by mentoring them on testing procedures, tooling, and the team's diagnostic and HIL workflows.

California State University, Fullerton

Fullerton, CA

Research Assistant — Hardware Security Lab

Aug 2025 – May 2026

- Identified a racing-path glitch failure mode in an Anderson PUF design through simulation, then attempted direct hardware capture with a logic analyzer and oscilloscope and determined the glitch exceeded available instrument sampling bandwidth, establishing placement sensitivity as a documented (published, [IEEE ISQED 2026](#)).

- Eliminated 90% of manual placement effort by automating Vivado placement, synthesis, and bitstream generation with TCL, mapping 128 PUF instances to precise SLICE locations via parsed CSV placement constraints.
- Cut PUF data collection and processing from two weeks of manual labor to one automated week across 2,556 configurations by building an end-to-end [Python GUI pipeline](#) for RealTerm instrument control, raw response acquisition, data cleaning, and majority-voting metric computation.
- Designed a library of functionally diverse Hardware Trojans in Verilog and diagnosed synthesis-induced Trojan erasure in Vivado, iterating on RTL structure to guarantee Trojan persistence in the implemented netlist prior to trace acquisition ([IEEE HOST 2026 Hardware Demo](#)-Demo ID: 243).

California State University, Fullerton

Fullerton, CA

Teaching Assistant — Tutor and Grader, EGE303 Electronics

Spring 2024 – Fall 2025

- Tutored and graded 50+ undergraduates in electronics coursework (MOSFETs, BJTs, op-amps, small-signal analysis), focusing on circuit debugging and schematic interpretation while coordinating with faculty to ensure consistent evaluation.

PROJECTS

Custom IR Sensor PCB — Line-Following Robot (Altium Designer)

- Designed the schematic and PCB layout for a custom IR sensor array, calculated resistor and component values from datasheets, then fabricated, soldered, and integrated the board onto the vehicle.

RTOS Motor Speed Controller — Tiva-C TM4C123 (ARM Cortex-M4)

- Built a bare-metal cooperative/blocking RTOS kernel supporting round-robin and priority scheduling, then implemented interrupt-driven motor speed control and SPI-driven LCD updates with all drivers written at the register level.

EDUCATION

M.S., Computer Engineering — California State University, Fullerton

Fall 2023 – May 2026

- GPA: 3.87/4.0 — Tau Beta Pi Engineering Honor Society (Member)